

Prince Sultan University – College of Engineering
Department of Communications and Networks Engineering
Electrical Engineering Program
Course Syllabi

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- 2 - MATH 113 CALCULUS II
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1. **Course number and name:** MATH 111 Calculus I
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Prof. Kamaleldin Abodayeh
4. **Textbook:** Calculus: Early Transcendent Functions. 9th Edition, by J. Stewart, D Clegg, S. Watson; 2021
5. **Course Information**
 - a. **Description:** This course is designed to develop the topics of differential calculus. Emphasis is placed on limits, continuity and derivatives of algebraic and transcendental functions of one variable. The goal of this course is developing the student's geometric insight into the concepts of differentiation, and applying these concepts to problem solving and "real world application"
 - b. **Prerequisite or Co-requisite:** NA
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Determine whether a function is continuous and/or differentiable	1
	Solve various real-world problems using calculus concepts	1
	Calculate the limit of a function using appropriate techniques including L'Hopital's rule	1
Skills	Compute the derivative of functions by applying the rules of differentiation	6
	Apply the concept of differentiation to sketch the graph of a function in a systematic way	6
	Find extreme values of functions	6
Values		

7. **Topics to be Covered:**
 - Introduction to functions, The Limit of a Function, Calculating Limits Using the Limit Laws
 - Continuity Limits at Infinity; Horizontal Asymptotes
 - Derivatives and Rate of Change, Derivative of Polynomials and Exponential Functions

- The Product and Quotient Rules, Derivatives of Trigonometric Functions, The Chain Rule
- Derivatives of Logarithmic Functions, Related Rates, Maximum and Minimum Values
- The Mean Value Theorem, How Derivatives affect the Shape of a Graph
- Indeterminate Forms and L'Hopital's Rule
- Summary of Curve Sketching
- Optimization

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1. **Course number and name:** MATH 113 Calculus II
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Prof. Aiman Mukheimer
4. **Textbook:** Calculus: Early Transcendental Functions. International Metric Edition 9th edition
By Stewart, 2021
5. **Course Information**
 - a. **Description:** This course introduces various topics such as: the concept of anti-derivatives, definite and indefinite integrals, the fundamental theorem of Calculus, techniques of integration and improper integrals. Applications of definite integrals to find area, volume, arc length and surface of revolution. Sequences and infinite series: convergence tests, integral and comparison tests, ratio and root tests, alternating series and Power series.
 - b. **Prerequisite or Co-requisite:** Math111
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Evaluate the integration of basic functions such as polynomials, exponential function, logarithmic functions, etc. by using the definition of integration, anti-derivative and standard rules.	1
	Find the area of plane region, arc length and surface of revolution by using integration techniques.	1
	Use the integration methods to find the volume of solid generated by rotating region about vertical axis or horizontal axis	1
Skills	Evaluate the integration of complicated functions by using methods of integration such as substitution, integration by parts, Trigonometric substitutions, partial fractions and trigonometric integrals.	6
	Determine the convergence and divergence of sequences and series	6
Values		

7. **Topics to be Covered:**

- Ch.5 Integrals– CLO1
- Ch.6 Applications of Integration– CLO2 and CLO3
- Ch.7 Techniques of Integration– CLO4
- Ch.8 Further Applications of Integration– CLO2
- Ch.11 Sequences, Series, and Power Series– CLO5

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1. **Course number and name:** MATH 215 Engineering Mathematics
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Walid Dyab
4. **Textbook:** Engineering Electromagnetics, Nathan Ida, 2015, Springer

5. **Course Information**

- a. **Description:** This course provides a solid foundation in multivariate calculus and complex variables essential for engineering applications. The topics covered in this course are functions of several variables, double integrals and triple integrals in rectangular, cylindrical, and spherical coordinates. The course also covers elementary topics in complex variables, including their properties, operations with complex numbers, polar and rectangular forms, phasors, and graphical representation
 - b. **Prerequisite or Co-requisite:** MATH 113
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding		
Skills	Calculate limits and partial derivatives for functions of 2 and 3 variables.	2
	Solve optimization problems in functions of 2 variables.	3
	Evaluate double and triple integrals in different coordinate systems for area and volume.	6
	Analyze vectors and linear transformations in Euclidean inner product spaces geometrically and algebraically.	2
	Determine the dimension and orthonormal bases for vector spaces.	6
Values		

7. **Topics to be Covered:**

- Introduction to Scalars and Vectors, magnitude and Direction of Vectors: The Unit Vector and Components of a Vector, Vector Addition and Subtraction, and Vector Scaling.

- Products of Vectors, the Scalar Product, the Vector Product. Multiple Vector and Scalar Products
- Definition of Fields, Scalar Fields, Vector Fields
- Systems of Coordinates, The Cartesian Coordinate System, The Cylindrical Coordinate System, The Spherical Coordinate System, Transformation from Cylindrical to Spherical Coordinates, and Position Vectors.
- Introduction to Vector Calculus, Integration of Scalar and Vector Functions Line Integrals, Surface Integrals, Volume Integrals
- Differentiation of Scalar and Vector Functions, The Gradient of a Scalar Function, The Divergence of a Vector Field, The Divergence Theorem, Circulation of a Vector and the Curl, Stokes' Theorem, and Conservative and Nonconservative Fields
- Null Vector Identities and Classification of Vector Fields, The Helmholtz Theorem, Second-Order Operators, Other Vector Identities
- Complex number, polar and rectangular forms of complex numbers, analysis of functions of complex variables

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1. **Course number and name:** MATH 225 Differential Equations
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Jehad Alzabut
4. **Textbook:** Differential Equations with Boundary Value Problems Author: D. G. Zill, W. S. Wright. Edition: 10th, 2024. Publisher: Cengage

5. **Course Information**

- a. **Description:** This course introduces the students to various topics in ordinary differential equations. These topics include: techniques for solving first order differential equations such as linear equations, separable equations, exact equations and integrating factors; homogeneous and general second order linear equations; higher order linear equations; power series solutions; the Laplace transform and applications in science and engineering, Fourier series and its convergence theorem, Separation of Variables, Heat Equation, Wave Equation and Laplace equation.
 - b. **Prerequisite or Co-requisite:** MATH 113
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
- Instructional Outcomes and their mapping to Student Outcomes:**

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize the characteristics of various types of differential equations	1
	Apply analytical techniques to solve 1-st, 2-nd and higher order differential equations	1
	Analyze to demonstrate models of differential equation in fields such as physics, biology and engineering	1
Skills		
Values		

7. **Topics to be Covered:**

- Initial value problem, Separable equations
- Linear equations

- Reduction of order, Homog. Linear equations with constant coefficients
- Method of Undetermined Coefficient-Superposition Approach.
- Cauchy Euler equation.
- Review on power series.
- Definition of Laplace transforms.
- Orthogonal functions, Fourier series. Fourier cosine and sine series.
- Separable PDE, Classical PDE and BVP.

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1. **Course number and name:** MATH 223 Linear Algebra
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Bahaaeldin Abdalla
4. **Textbook:** Elementary Linear Algebra by Anton, Rorres & Kaul, 12th Edition
5. **Course Information**
 - a. **Description:** The course examines the techniques of linear algebra and utilizes its tools. Topics and instruments include matrices, determinants, systems of linear equations, Euclidean vector spaces, real vector spaces, inner product spaces, eigenvalues and eigenvectors, linear transformation, applications
 - b. **Prerequisite or Co-requisite:** MATH 113
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Identify matrices, determinants, and their vector operations	1
	Grasp the concepts of vector spaces, subspaces, and linear transformations between vector spaces.	1
	Determine and obtain orthogonality in inner product spaces	1
Skills	Perform matrix operations to solve linear systems and to determine linear independence of vectors and bases of subspaces	2
	Compute eigenvectors and eigenvalues of matrices and use them in the diagonalization process.	6
Values		

7. **Topics to be Covered:**
 - Systems of Linear Equations, Gaussian Elimination, Matrices and Matrix Operations, Invertible, Diagonal, Triangular, and Symmetric Matrices, Compositions of Matrix Transformations.
 - Determinants by Cofactor Expansion, Determinants by Row Reduction, Properties of Determinants, Cramer's Rule.

- Vectors in n -Space, Norm, Dot Product, and Distance in
- n , Orthogonality, The Geometry of Linear Systems, Cross Product.
- Real Vector Spaces, Subspaces, Spanning Sets, Linear Independence, Coordinates and Basis, Dimension, Change of Basis.
- Eigenvalues and Eigenvectors, Diagonalization.
- Inner Products, Angle and Orthogonality in Inner Product Spaces, Gram–Schmidt Process, Best Approximation; Least squares, Mathematical Modeling Using Least Squares.
- Orthogonal Matrices, Orthogonal Diagonalization.
- General Linear Transformations, Compositions and Inverse Transformations.

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1. **Course number and name:** PHY 105 Physics I
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Hazem Abu-Farsakh
4. **Textbook:** Physics Technology Update, by James S. Walker. Fourth Edition, Pearson International Edition, 2014
5. **Course Information**
 - a. **Description:** This course introduces the principles of mechanics, energy, and heat. The course covers physics and measurement, motion in one dimension, vectors, motion in two dimensions, laws of motion, circular motion and other applications of Newton's laws, work and energy, potential energy and conservation of energy, rotational kinematics and torque, temperature, and heat. The emphasis in this course is cultivating an understanding of natural phenomena through direct observation, reasoning, and application of this knowledge.
 - b. **Prerequisite or Co-requisite:** NA
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe basic physical quantities in mechanics and heat.	1
	Calculate various physical quantities using translational and rotational kinematical equations and torque	1
	Apply Newton's laws of motion to solve various physical problems.	1
Skills	Solve physical problems using conservation laws.	2
	Calculate physical parameters associated with the thermal properties of matter	2
	Demonstrate basic experimental skills by setting up the experiment, carrying out the experimental procedure, and report results properly	6
Values		

7. **Topics to be Covered:**

- Ch. 1: Introduction to Physics. Physics and the laws of Nature, Units of Mass, Length and Time, Dimensional analysis, Converting Units
- Ch. 2: One Dimensional Kinematics. Position, Distance and Displacement, Average Speed and Velocity, Acceleration, Motion with Constant Acceleration, Freely Falling Objects
- Ch. 3: Vectors. Scalars and Vectors, Components of a Vector, Adding and Subtracting vectors, Unit Vectors, Position, Displacement, Velocity, and Acceleration Vectors.
- Ch. 4: Two-dimensional kinematics. Projectile Motion, Zero Launch Angle, General Launch Angle, Key Characteristics.
- Ch. 5: Newton's Laws of Motion. Force and Mass, Newton's First Law of Motion, Newton's Second Law of Motion, Newton's Third law of Motion, Weight, Normal Forces
- Ch. 6: Application of Newton's Laws. Frictional Forces, Strings and Springs, Translational Equilibrium, Circular Motion.
- Ch. 7: Work and Kinetic Energy. Work done by a constant Force, Kinetic Energy and Work-Energy Theorem, Work done by a variable force (spring), Power
- Ch. 8: Potential Energy and Conservation of Energy. Potential Energy and the Work done by Conservative forces, Conservation of Mechanical Energy.
- Ch. 9: Linear Momentum and Collision. Linear Momentum, Momentum and Newton's Second law, Impulse, Conservation of Linear Momentum, Inelastic Collision, Elastic Collision

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1. **Course number and name:** PHY 205 Physics II
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Pr. Mohamed Bououdina
4. **Textbook:** Physics, Technology Update, Fourth Edition, James S. Walker, Pearson.
5. **Course Information**
 - a. **Description:** This course includes several topics in electricity, magnetism, and optics. Topics include: The electrostatic force, electric field, Gauss's law, electric potential, capacitors and dielectrics, direct current circuits, magnetic fields, electromagnetism, induction, Faraday's law, alternating current circuits, geometrical optics, and optical instruments.
 - b. **Prerequisite or Co-requisite:** PHY 105
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe basic concepts of electricity, magnetism, and optics	1
	Describe the image formation and its application in optical instrument	1
Skills	Calculate electric forces, fields and potentials associated with different charge arrangements	6
	Analyze electric circuits to determine unknown variables.	6
	Apply the knowledge of magnetic forces and fields to relevant problems	6
	Demonstrate basic experimental skills by setting up laboratory equipment safely, carrying out experimental procedure, and reporting the results of the experiment	6
Values		

7. **Topics to be Covered:**
 - Electric charge, Insulators and conductors, Coulombs' Law. The electric field lines and electric fields due to symmetric charge configurations. Induction. Electric flux and Gauss' Law.

- Electric potential Energy and Electric potential. Energy conservation. The electric potential due to a point charge and other symmetrical charge configurations. Equipotential surfaces and the electric field. Capacitors and dielectrics. Electrical energy storage.
- Electric current. Resistance and Ohms' Law. Energy and Power in Electric Circuits. Resistors in series and in parallel. Kirchhoff's Rules. Capacitors circuits. RC circuits.
- The magnetic fields. The Magnetic force exerted on a current carrying conductor. The magnetic force on moving charges. Electric currents, Magnetic fields, and Amperes' Law. Current loops, Solenoid and Toroid.
- Induced Electromotive force. Magnetic Flux. Faradays' Law of Induction.
- The reflection of Light. Image formation by plane mirror. Spherical mirrors. The Mirror equation and Image formation. Magnification.

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1. **Course number and name:** CHM 101 General Chemistry
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Ihab Shawish
4. **Textbook:** Chemistry 10th edition by Steven Zumdahl and Susan Zumdahl; ISBN:ISBN-13: 978-1305957404, ISBN-10: 9781305957404, 2016

5. **Course Information**

- a. **Description:** This course is designed to provide students with foundational knowledge of chemistry. The course content encompasses basic concepts, including units and measurements, the atomic theory, principles of stoichiometry, properties of aqueous solutions, behaviors of gasses, thermochemistry, quantum numbers, periodic trends, chemical bonding, VSEPR theory, and the classification of organic compounds
 - b. **Prerequisite or Co-requisite:** NA
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
- Instructional Outcomes and their mapping to Student Outcomes:**

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize the basic concepts, theoretical fundamentals, and the related experimental applications in chemistry	1
	Differentiate between the major classes of organic compounds and understand the IUPAC nomenclature of simple organic compounds	1
	Realize the main principles of thermochemistry and gases	1
Skills	Predict the solution of the problem, apply suitable strategy to reach the solution, test the correctness of the solution, and interpret the results	6
	Apply the principles of stoichiometry in different types of chemical reactions including reactions of aqueous solutions	6
	Interpret periodicity concepts, distinguish the types of chemical bonding, estimate the Lewis structure and molecular geometry of different chemical compounds.	6
Values	Use different experimental techniques and work within a group in the laboratory, understand the objectives of the experiments, analyze data, and interpret results for deep understanding of the theoretical topics.	5

7. Topics to be Covered:

- Ch. 1: Introduction, Chemistry an Overview, Scientific Method, Units of Measurements, Uncertainty in Measurements, and significant figures.
- Ch.2: Fundamental Chemical Laws, the Modern View of Atomic structure: An introduction, Molecules and Ions, The Periodic Table, Naming Simple Compounds.
- Ch.3: Atomic Masses, The Mole, Molar Masses, Learning to Solve Problems, Percent Composition of Compounds, Determining the Formula of Compounds , Chemical Equations, Balancing Chemical Equations, Stoichiometric Calculations, The concept of Limiting Reactant.
- Ch. 4: Water the common solvent, The nature of aqueous solutions, The composition of solutions: Molarity, Types of Chemical Reactions, Precipitation Reactions, Describing Reactions in Solutions, Stoichiometry of Precipitation Reactions, Acid Base Reactions, Oxidation- reduction Reactions
- Ch. 5 Gases: Pressure, The gas laws of Boyle, Charles, and Avogadro, The Ideal Gas Law, Gas Stoichiometry, Dalton's Law of Partial Pressures: Gas Mixtures
- Ch. 6 Thermochemiostry: The Nature of Energy, Enthalpy and Calorimetry, Hess's Law, Standard Enthalpies of Formation
- Ch. 7: Quantum Numbers, Orbitals Shapes and Energies, Electron Spin and Pauli Principle, The Aufbau Principle, Periodic Trends in Atomic Properties.
- Ch. 8: Types of Chemical Bonds, Electronegativity, Polarity, Drawing Lewis Structure, Exceptions of Octet Rule, Resonance, Molecular geometry and VSEPR Theory
- Ch. 22 Organic Chemistry: Classes of Organic Compounds, Nomenclature of Organic Compounds, Addition and Substitution Reactions

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1. **Course number and name:** STAT 101 Introduction to Statistics & Probability Theory
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Ateeq ur Rehman Irshad
4. **Textbook:** Business Statistics by Groebner, Shannon, Fry. Tenth Edition

5. **Course Information**

- a. **Description:** Introduction to Statistics & Probability Theory is the first course in Statistics, designed to expose the students to the use of tables and graphs as a way to summarize data and provide concise and meaningful presentations for the data. Using the computing facilities at campus, problem-solving is spreadsheet-oriented whenever possible. Excel is a spreadsheet tool for manipulating quantitative data. The extent to which the variables are related is discussed deeply through graphs and using rules. The emphasis is on analyzing, understanding, and using statistics to solve real-world problems. Coverage of the course includes descriptive statistics, probability and probability distributions, and sampling.
 - b. **Prerequisite or Co-requisite:** MATH 111
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
- Instructional Outcomes and their mapping to Student Outcomes:**

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe Data, Population, sample, parameters, statistic and variables	1
	Explain the concepts of descriptive statistics and probability rules	1
	Calculate the central tendencies, location, dispersion, Discreate and continuous probability distributions.	1
Skills	Illustrate the data through suitable statistics and graphical display.	3
	Formulate a research study through survey and group projects using computer skills.	3
Values		

7. **Topics to be Covered:**

- The Where, Why, and How of Data Collection
- Graphs, Charts, and Tables—Describing Your Data

- Describing Data Using Numerical Measures
- Introduction to Probability
- Discrete Probability Distributions
- Introduction to Continuous Probability Distributions
- Introduction to Sampling Distributions
- Estimating Single Population Parameters

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1. **Course number and name:** ENG 301 English Writing and Translation
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Reem Alsufayan
4. **Textbook:** Technical Communication 13th ed., Mike Martel; Stuart A. Selber Macmillan International, Higher Education, 2021 by Bedford/St. Martin's
5. **Course Information**
 - a. **Description:** This course introduces the principles and practices of technical writing. The object of the course is cultivating writing skills cued to professional work. Through an array of homework assignments, engaged class discussions and report projects, students obtain proficiency in various aspects of technical writing. Attention is also given to the development of argumentative and analytical skills as well as elements of document design and formatting.
 - b. **Prerequisite or Co-requisite:** ENG 101
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Demonstrate knowledge of effective technical correspondence with an awareness of audience, purpose, and context.	1
Skills	Write technical proposals/long reports based on field-specific research.	3
	Demonstrate knowledge of effective technical correspondence with an awareness of audience, purpose, and context.	3
	Evaluate the effectiveness of technical documents.	3
	Write clear and concise technical correspondence.	
Values	Demonstrate ethical communication and integrity in group projects.	4

7. **Topics to be Covered:**
 - Introduction to Technical Writing: What is technical Communication? Why technical writing skills are important in your career.
 - Characteristics of a Technical Document

- The importance of Audience: Using an audience profile sheet, determining the important characteristics of your audience, applying what you have learned about your audience, writing for multiple audiences, and determining your purpose
- Writing Correspondence: Understanding the process of writing correspondence, selecting a type of Correspondence, presenting yourself effectively in correspondence, writing Letters, writing Memos, and writing Emails
- Applying for a job: Establishing your professional brand, finding the right position, writing resumes, and writing job application letters.

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1. **Course number and name:** EE 101 Computer Programming for Engineering
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Muddesar Iqbal
4. **Textbook:** E Balagurusamy, "Problem Solving and Python Programming", McGraw Hill Education, 2018.
5. **Course Information**
 - a. **Description:** This course covers the concept of programming languages in Python. It enables the student to understand and write basic coding based on various operators, functions, logic statements, Strings, tuples, etc., in the Python programming language. In addition, the course covers the basic concept of files and exceptions.
 - b. **Prerequisite or Co-requisite:** NA
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Define fundamental concepts of programming along with problem-solving techniques and analytical thinking.	1
Skills	Explain the various logic using various operators and expressions.	3
	Write a programming code in Python.	2
	Examine program code using Python	6
Values		

7. **Topics to be Covered:**
 - Introduction to Digital Computers and Problem-Solving Strategies.
 - Introduction to Python and Its Importance; Getting Started with Python: Input/Output Statements, Basic Structure of Python Programs, Programming Style, Compilation, and Execution.
 - Keywords and Identifiers, Constants, Variables, Data Types, and Python Instructions.
 - Declaration, Assigning Values, Defining Variables, Operators, and Expressions.

- Managing Input and Output Operations; Decision Making and Branching, Decision Making and Looping.
- Python Strings, Lists, Tuples, and Dictionaries.
- Introduction to Microprocessors; Setting Up a Microprocessor (e.g., Raspberry Pi or a Simulator) to Run Python Code and Create Python Projects.
- Functions: Built-in Functions, User-Defined Functions, Parameters and Arguments, Function Calls, Return Statements, Python Recursive Functions, Anonymous Functions.
- Files and Exceptions: Text Files, Directories, Exceptions, Exceptions with Arguments, and User-Defined Exceptions.

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1. **Course number and name:** EE 201 Introduction to Circuits
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Maged Esmail
4. **Textbook:** Robert L. Boylestad - Introductory Circuit Analysis (2016, Pearson), 13th Edition

5. **Course Information**

- a. **Description:** This module provides a comprehensive introduction to both DC and AC circuit principles. Students will begin by reviewing fundamental concepts, including current, voltage, and power. The course then explores series and parallel circuits and delves into Ohm's Law, Kirchhoff's Laws, source transformations, bridge networks, and delta/star transformations. Key analysis methods and network theorems are also covered. Additionally, the module revisits inductors and capacitors, introduces sinusoidal sources and the concept of phasors, and details methods for analyzing AC circuits. Practical experience is emphasized through hands-on use of electrical measurement tools such as voltmeters, ammeters, wattmeters, function generators, and oscilloscopes.
 - b. **Prerequisite or Co-requisite:** PHYS 205, Math113
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
- Instructional Outcomes and their mapping to Student Outcomes:**

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize the main concepts of electrical parameters and their interrelations.	1
Skills	Analyze DC/AC electric circuits using Ohm's law, Kirchhoff's circuit laws, and voltage/current divider rules.	6
	Analyze DC/AC electric circuits using different analysis techniques such as Mesh, Nodal, and Thevenin.	6
	Analyze sinusoidal signals, reactance, admittance, and power using phasor forms.	6
Values	Demonstrate DC/AC electric circuits using hardware boards and simulation in group activities.	5

7. Topics to be Covered:

- Introduction, Basic concepts, charge, current, voltage, power, energy, resistance, and Ohm's Law, dissipated power.
- Introduction, Series, Parallel and Series parallel circuits, Kirchhoff's laws, voltage division and current distribution principles, voltage and current sources, open and short circuits
- Mesh and Nodal Analysis, Bridge networks, Y- Δ and Δ -Y conversion
- Superposition, Thevenin's and Norton's theorems
- Review of capacitance and inductance, Generation of AC voltages, parameters of AC. waveforms, average and effective (RMS) values
- Review of complex number algebra, phasor representation, impedance and admittance. Capacitive and inductive reactance
- Examples of analysis methods for AC circuits
- General equation, Resistive circuit, Apparent Power, Inductive circuit and reactive power, Capacitive circuit, The power triangle, The total P, Q, and Power factor correction

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1. **Course number and name:** EE 211 Electronic Fundamentals
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Moustafa Nasralla
4. **Textbook:** Electronic Devices (Conventional Current Version): Pearson New International Edition, Author(s): Floyd, Thomas L., Publisher: Pearson (Intl), Edition: 9th, Pages: 952, Print ISBN: 9781292025643, 1292025646eText ISBN: 9781292038070

5. **Course Information**

- a. **Description:** This course provides a comprehensive introduction to essential semiconductor devices and amplifiers. It covers diodes, including special-purpose types, and explores Bipolar Junction Transistors (BJTs) with a focus on their operation, characteristics, and various biasing methods. Students will study transistor amplifier configurations such as common emitter, common collector, and common base, as well as classes of power amplifiers. The course also delves into operational amplifiers (Op-Amps), addressing their parameters, feedback mechanisms, and various practical circuit applications.
- b. **Prerequisite or Co-requisite:** EE 201
- c. This is a Required course.

6. **Goals:** Specific goals for the course

Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe the basic electronic circuits using diodes and transistors.	1
Skills	Analyze the characteristics of different BJT configurations and biasing circuits.	6
	Analyze operational amplifier characteristics and circuits for different applications.	6
Values	Demonstrate experimentally basic electronic circuits and their applications	5

7. **Topics to be Covered:**

- Introduction Diodes and Applications, Special purpose diodes
- Intro Bipolar Junction Transistor – BJT, BJT Operation
- Stabilization, Voltage divider bias method
-
- Ch/cs curves, modes of operation, load lines, Ratings, Q-Point, BJT applications
- Basic of Amplifier AC parameters - Transistor AC Models
- Classes of Power amplifiers
- Introduction to Operational Amplifiers Parameters
- Op-Amps: Inverting and non-inverting amplifier,
- The Common-Collector Amplifier

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 221 Logic Design
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Walid Dyab
4. **Textbook:** Digital Systems: Principles and Applications, 12/E, 2017, Neal Widmer, Greg Moss, Ronald J. Tocci. ISBN-10: 0135103827, ISBN 13: 978-0-13-422013-0, Publisher: Prentice Hall
5. **Course Information**
 - a. **Description:** This module enables students to understand concepts in binary numbers, number base conversion, complements and codes, definition of Boolean Algebra, Boolean functions, digital logic gates, integrated circuits, Karnaugh map methods, combinational logic circuits, sequential logic circuits and Memory modules. Design and analysis of sequential logic circuits such as: Shift Registers, Counters, Synchronous and Asynchronous Sequential Circuits, State Diagrams, State Tables, Students will be able to develop, measure, and test different types of Sequential Circuits using D-type, T-type, and JK-type Flip-Flops.
 - b. **Prerequisite or Co-requisite:** EE 101
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe integrated-circuit logic families, memory devices, and programmable logic devices.	1
Skills	Explain different binary number systems and arithmetic.	6
	Explain Boolean expressions using Karnaugh Map.	6
	Analyze and design combinational logic circuit.	2
	Analyze and design sequential logic circuits including flip flops, counters, and registers.	2
Values	Demonstrate the ability to work in a team to collaborate in performing LAB experiments and projects.	5
	Illustrate the ability to use Multisim and FPGA boards in group activities through lab and projects.	4, 7

7. **Topics to be Covered:**

- Introductory Concept
- Number Systems and Codes
- Describing Logic Circuits
- Combinational Logic Circuits
- Flip-flops and Related Devices (Part 1,2)
- Flip-flops and Related Devices (Part 3)
- Digital Arithmetic operations
- Digital Arithmetic Circuits: (Binary Adders/ Subtractors/ Multipliers)
- Counters, registers

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 231 Signals and Systems
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Mourad Rizk
4. **Textbook:** Signals, Systems, & Transforms (5th Global Edition), Phillips, Charles; Parr, John; Riskin, Published by Pearson Education (2014). ISBN: 9781292015286
5. **Course Information**
 - a. **Description:** This course covers basic concepts and methods related to continuous and discrete-time signals and systems. The course includes: signals and systems and their properties, linear time-invariant systems, stability analysis, sampling of continuous-time signals, z-transform, continuous and discrete-time Fourier transform, discrete Fourier transform and fast Fourier Transform.
 - b. **Prerequisite or Co-requisite:** MATH225, EE 201
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe different properties of continuous and discrete-time signals and systems.	1
	Recognize the sampling theory.	1
Skills	Analyze continuous and discrete linear time-invariant (LTI) systems.	6
	Explain Z- transform and its applications for LTI systems.	3
	Explain Continuous and Discrete-time Fourier Transform and its applications for LTI systems	3
	Write MatLab code and sampling theory and develop discrete system.	2
Values		

7. **Topics to be Covered:**
 - Introduction to Signals and Systems
 - Continuous-Time Signals and Systems
 - Continuous-Time Linear Time-Invariant Systems
 - Continuous-time Fourier Transform

- Discrete-Time Linear Time-Invariant Systems
- Discrete-Time Fourier Transform
- The z-Transform

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1. **Course number and name:** EE 202 Circuit Analysis
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Mahagan Sagar Bhaskar
4. **Textbook:** Ravish R. Singh, "Electrical Networks" Tata McGraw-Hill Education, July 2009.

5. **Course Information**

- a. **Description:** The course begins by reviewing the behaviors of inductor and capacitor storing elements. It then covers the characteristics and behavior of first-order RL/RC circuits, as well as second-order RLC circuits. Additionally, the course explains the conversion of time-domain circuits to the s-domain using the Laplace transform. Furthermore, it delves into frequency response and resonance, providing a basic understanding of filter circuits. The course also includes the study of electrical configurations involving three phases of networks and the analysis of two-port networks.
 - b. **Prerequisite or Co-requisite:** EE 201, MATH 225
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
- Instructional Outcomes and their mapping to Student Outcomes:**

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize the application of the basic electric circuit components in different electrical applications.	1
Skills	Analyze first-order and second-order transient response of electrical circuits.	6
	Apply the Laplace transform in circuit analysis.	6
	Analyze frequency selective circuits	6
	Analyze three-phase networks and two-port networks.	6
Values		

7. **Topics to be Covered:**

- Review of circuit elements, first-order RL and RC circuits
- Second-order RLC circuits
- Laplace transform and its application in circuit analysis

- Frequency selective circuits and active filter circuits
- Single Phase and Three Phase System
- Two-port networks

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Prince Sultan University – College of Engineering
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1. **Course number and name:** EE 312 Electronic Engineering
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Mahagan Sagar Bhaskar
4. **Textbook:** Robert L. Boylestad Louis Nashelsky, ‘Electronic Devices and Circuit Theory’, Eleventh Edition, Pearson, 2013.
5. **Course Information**
 - a. **Description:** This course enables the students to understand the DC and AC analysis, and modelling of multistage configuration based on BJT, FET, MOSFET. This course also helps to analyze and design the different electronic circuitries- CMOS circuit such as NAND Gates, Transmission Gates and memory elements.
 - b. **Prerequisite or Co-requisite:** EE 211
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Identify the feature of various amplifier circuits, active filters, and oscillators.	1
Skills	Analyze the model of BJT amplifiers and their frequency response.	6
	Analyze the DC and AC model of various FET amplifiers.	6
	Design CMOS circuit such as NAND Gates, Transmission Gates and memory elements	2
Values		

7. **Topics to be Covered:**
 - Multistage BJT Amplifier:
 - Design and analysis of various multistage BJT Amplifiers, ‘re’ modelling of the various multistage BJT Amplifiers, BJT Frequency Response.
 - Field-Effect Transistors and FET Biasing:

- Introduction, Construction and Characteristics of JFETs, Transfer Characteristics, JFET versus BJT, Depletion-Type MOSFET, Enhancement-Type MOSFET, VMOS and UMOS Power and MOSFETs, CMOS, MESFETs.
- DC modelling of the FET, MOSFET configurations, Combination Networks, p-Channel FETs.
- FET AC Modelling and Amplifier
- Introduction, JFET Small-Signal Model
- Fixed-Bias, Self-Bias, Voltage-Divider, Common-Gate, Source-Follower (Common -Drain) Configurations,
- Depletion-Type MOSFETs, Enhancement-Type MOSFETs, E-MOSFET Drain-Feedback Configuration, E-MOSFET Voltage-Divider Configuration, Designing FET Amplifier Networks. Effect of R_L and R_{sig} , Cascade Configuration, Practical Applications.

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 322 Microprocessors Design
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Walid Dyab
4. **Textbook:** David Harris and Sarah Harris, “Digital Design and Computer Architecture” RISC-V Edition, Elsevier 2021
5. **Course Information**
 - a. **Description:** The course will focus primarily on basic embedded system concepts and the student will learn the basics of designing, interfacing, configuring, and programming embedded systems. One of popular microcontroller is selected to implement the techniques learned in class. By the end of the course the student will have mastered the basics of embedded system design and programming.
 - b. **Prerequisite or Co-requisite:** EE 211, EE 221, CEE 205
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Describe the types and operation of a typical microprocessors/microcontrollers.	1
	Recognize various memory technologies, cache memories, addressing modes, memory mapping, and composing memories.	1
Skills	Explain the operations of peripherals typically used for interfacing microprocessors, such as A/D & D/A converters, Timer, Counters, LCDs, UART, PWM controllers, regular and stepper motors, Real time clocks.	3
	Illustrate microprocessor communication basics, hardware interfacing techniques, I/O addressing, interrupts, direct memory access, arbitration, and multi-level bus architectures.	2
Values	Demonstrate the ability to configure hardware and software microprocessor-based systems	4, 7

7. Topics to be Covered:

- Introduction to digital electronics
- Number Systems and representation
- Digital Integrated Circuits - Memory
- Hardware Programming
- Computer Architecture and Microcontrollers
- Software programming and assembly

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 341 Electromagnetics Fundamental
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Maged Esmail
4. **Textbook:** W.H. Hayt, Jr. and J.A. Buck, “Engineering Electromagnetics,” 9th edition, McGraw Hill, 2018
5. **Course Information**
 - a. **Description:** This module is designed to introduce the electromagnetic fields. Coulomb's law, Gauss's law, electrical potential, dielectric materials capacitance, boundary value problems, Biot-Savart law, Ampere's law, Lorentz force equation, magnetic materials, magnetic inductance, time varying fields and Maxwell's equations
 - b. **Prerequisite or Co-requisite:** MATH 215, MATH 225, EE 201
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Explain the concept of electric field and magnetic field.	1
Skills	Analyze Maxwell’s equations to find the electric and magnetic field components in different media.	6
	Develop an ability to apply Gauss’ law, Ampere’s Law, Biot-Savart law, Faraday’s law and Maxwell’s equations in the analysis of electromagnetic systems.	6
	Calculate the percentage of electromagnetic energy reflected and transmitted from a medium to another.	2
Values		

7. **Topics to be Covered:**
 - Introduction & Review Coordinate Systems
 - Coulomb’s Law, Electric Field intensity,
 - Electric Flux Density,

- Gauss' Law, Divergence and first Maxwell equation
- Electric Potential, Elec. Dipole, Energy density
- Currents and current density
- Boundary Conditions (BC), Capacitance
- The Steady Magnetic Field: BIOT-SAVART LAW, STOKES' THEOREM
- Ampere's Laws, Mag. Flux Density Mag. Potential, Magnetic Forces and Magnetization, BC

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 332 Control Systems
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Dhafer Almakhlles
4. **Textbook:** Modern Control Modern Control Engineering, (5th Edition) By: Katsuhiko Ogata.
5. **Course Information**
 - a. **Description:** Introduction to feedback control systems. Block diagram and signal flow graph representation. Mathematical modeling of physical systems. Stability of linear control systems. Time-domain and frequency-domain analysis tools and performance assessment. Lead and lag compensatory design. Proportional, integral, and derivative control.
 - b. **Prerequisite or Co-requisite:** EE 231, MATH 223
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize feedback in control systems and stability concept	1
Skills	Apply the Laplace Transform in modelling the LTI control systems	2
	Recognize feedback in control systems and stability concept	2
	Evaluate the behaviour of LTI in time domain	6
	Design a stable control system according to the required specification	
Values	Demonstrate the concepts of stable and well-performed control systems	7

7. **Topics to be Covered:**
 - Introduction to Control Systems and Modelling EE systems
 - Laplace transform and Block Diagram Representation of Control Systems
 - Time-domain analysis of control systems:
 - Time Domain Analysis of 1st Order Systems
 - Time Domain Analysis of 2nd Order Systems
 - Steady State Error & Routh-Herwitz Stability Criterion

- Frequency Response Analysis
- Design of Feedback Control Systems:
- Lead and lag compensatory design.

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 351 Communication Systems
2. **Credits and contact hours:** 4 (6)
3. **Instructor:** Dr. Mourad Rizk
4. **Textbook:** Lathi, Modern Digital and Analog Communication Systems, Fifth Edition, Oxford University Press, 2018.

5. **Course Information**

- a. **Description:** In this course, we focus on signal transforms and continuous-wave modulation, which is the basic operation of analog communication systems. Initially, we give the students an insight and understanding of signals classifications, Fourier series, Fourier transform, spectrum analysis, and explore their applications in the context of analog communication systems. We then cover thoroughly amplitude modulation including the generation and reception of double-side band, single side-band, and vestigial side-band. We also cover the angle modulations (phase and frequency), the generation both the narrow and wide-band angle modulation.
 - b. **Prerequisite or Co-requisite:** EE 211, EE 231
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
- Instructional Outcomes and their mapping to Student Outcomes:**

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize different analog modulation and demodulation techniques.	1
Skills	Explain the linear model of the distortionless channel.	3
	Recognize the basic model of communication systems, such as AM and FM radio systems.	2
	Analyze the impact of various channels on communication systems.	6
	Demonstrate the pulse modulation techniques.	3
Values	Design modulation and demodulation circuits using various lab experiments, such as: envelope detector, rectifier circuits, and software defined radio circuits.	5

7. **Topics to be Covered:**

- Introduction to communication systems, analog and digital messages, channel effect, modulation and detection
- Signals and Signal Space
- Signals classification, some important signals, Fourier Series, Fourier Transform, properties of Fourier transform
- Analysis and transmission of signals
- signal transmission, signal distortion, ideal versus practical filters, essential bandwidth
- Amplitude modulations and demodulations
- Baseband vs passband, double-sideband suppressed carrier modulation
- Ordinary AM modulation, single-sideband AM, Quadrature AM, vestigial sideband
- FDM, Synchronization, superheterodyne receiver

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Prince Sultan University – College of Engineering
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Course Syllabus

1. **Course number and name:** EE 305 Power Electronics

2. **Credits and contact hours:** 4 (6)

3. **Instructor:** Dr. Umashankar Subramaniam

4. **Textbook:** Rashid M.H., ‘Power Electronics Handbook’, Academic Press, 2001.

5. **Course Information**

- a. **Description:** This course is an introduction to Power Electronics. It enables the student to understand the basic Power Electronics components, operations, design, and the analysis of basic power electronics converter circuits. The course is focused on the applications of the power converter in various sectors, e.g., industrial, electric grid, renewable energy, microgrid, electric vehicle, etc. In addition, the course covers the basic characteristics, control schemes, and applications of power semiconductor devices and circuits.
- b. **Prerequisite or Co-requisite:** EE 211
- c. This is a Required course.

6. **Goals:** Specific goals for the course

Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Recognize the characteristics of various semiconductor devices and power converter configurations.	1
Skills	Analyze the various configurations of AC-DC converter.	2
	Analyze the various configurations of DC-DC converter.	2
	Analyze the various configurations of DC-AC converter.	2
	Analyze the various configurations of AC-AC converter.	2
Values	Demonstrate various power electronics circuits, application, and analyze the waveform through simulation and experimental work	4, 7

7. **Topics to be Covered:**

- Introduction of power semiconductor device, Structure, and operating characteristics of power, power BJT, MOSFET and IGBT, switching characteristics, firing circuits, basic concepts of PWM control.

- Single Phase AC to DC Controlled converter configurations – Semi-converter – Fully controlled converter – R, RL, RLE load – operation under continuous and discontinuous conduction – Firing Schemes for converter- Dual converters.
- Basic principles of DC-DC- Converter, step-down and step-up chopper-control strategy, Choppers-A, B, C, D and E, volt-second and ampere-second balance equations- Steady state analysis, CCM and DCM characteristics, Buck Converter, Boost Converter, and Buck - Boost converter, Cuk Converter. Numerical and analysis examples
- Single phase and three phase voltage source inverters (both 120o mode and 180o mode)– Voltage & harmonic control- PWM techniques: Multiple PWM, Sinusoidal PWM, modified sinusoidal PWM.
- Single phase and Three phase AC voltage controllers, Single phase and Three phase AC voltage cyclo-converter.
- Application of Power Electronics Converter in various energy sectors e.g. industrial, electric grid, renewable energy, microgrid, electric vehicle etc.

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Prince Sultan University – College of Engineering
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1. **Course number and name:** EE 304 Electrical Machines
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Mahagan Sagar Bhaskar
4. **Textbook:** BL Theraja, " A Textbook of Electrical Technology - Volume II", S. Chand,
5. 2023.
6. **Course Information**
 - a. **Description:** The students will learn the fundamentals of electrical machinery. They will learn to analyze the voltage-current, torque-speed characteristics of the different machines. The students also will be able to understand the transformers, principles, and operation
 - b. **Prerequisite or Co-requisite:** EE202, EE341
 - c. This is a Required course.
7. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Explain the concepts of electromechanical energy conversion systems	1
	Describe the construction, principle of operation, and characteristics of dc machines, ac machines and transformers	1
Skills	Analyze the selection, methods of speed control, and methods of direction reversal of and applications of d.c. machines	6
	Analyze the selection, methods of speed control, and methods of direction reversal of and applications of a.c. machines	6
Values	Show the importance of sustainability in electrical machines and transformers	4

8. **Topics to be Covered:**
 - Introduction to Magnetic circuit concepts
 - Electro-mechanical systems, Motor/load interactions
 - Transformer theory, EMF, OC, SC operation
 - DC Generators, Types and Characteristics

- DC machines, Types, Speed Control, and Characteristics
- Starting methods and Braking
- Induction Machines, Types, Speed Control
- Synchronous Machines and Special Machines

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Prince Sultan University – College of Engineering
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1. **Course number and name:** EE 403 Power Systems
2. **Credits and contact hours:** 3 (4)
3. **Instructor:** Dr. Umashankar Subramaniam
4. **Textbook:** D. P. Kothari, I. J. Nagrath, Power System Engineering, 2019, 3rd edition, McGraw-Hill Education
5. **Course Information**
 - a. **Description:** This course provides an introduction to power systems used in electrical engineering. Examples include different types of generation systems, load flow studies and infinite bus systems with exploring their applications in substations and power plants.
 - b. **Prerequisite or Co-requisite:** EE 304
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Explain the basic concepts, components, and principles of power systems	1
	Describe the different types of power generators and their characteristics	1
Skills	Analyse both symmetrical and unsymmetrical conditions in power systems.	2
	Analyze the behavior of power systems under steady-state and transient conditions	2
	Evaluate the per-unit system and draw the single-line diagram	6
Values		

7. **Topics to be Covered:**
 - Power and power factor calculations in single phase and three phase power
 - Generation of electrical energy (coal, hydro, gas, sun, water, etc.)
 - Network calculations for balanced and unbalanced, star and delta connected, three phase networks

- Load flow calculations
- The per unit system representation
- Single line equivalent diagrams
- power flow calculations

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Prince Sultan University – College of Engineering
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1. **Course number and name:** EE 490 Senior Design Project
2. **Credits and contact hours:** 3 (3)
3. **Instructor:** Dr. Moustafa Nasralla
4. **Textbook:** Exploring Engineering: An Introduction to Engineering and Design / Philip Kosky, ISBN 978-0-12-374723-5
5. **Course Information**
 - a. **Description:** Project groups deliver their products that have progressed through the design, analysis, testing and evaluation stages. The project teams produce a polished professional report that describes the design process, implementation and testing, verification and validation, and a critical appraisal of the project. An oral presentation and graphic works are complementary project deliverables
 - b. **Prerequisite or Co-requisite:** Completion of 90 Credit Hours, ENG 301, CEE 205
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Identify, gather and synthesize relevant information needed to define and solve an open-ended design problem.	1
Skills	Apply the engineering design process to define, formulate and solve a conceptual design problem considering human factors, safety, environmental, economic, and societal constraints.	2
	Use lateral and critical thinking to generate creative alternative concepts and identify a preferred solution to a design problem.	6
	Use engineering tools and drawing skills to check and visually communicate design problem-solving.	3
	Communicate effectively and define engineering design problems and keep a well build log file and keep a journal of ideas and activities associated with the design project.	3
Values	Recognize the engineering profession and the importance of ethical and legal factors in professional duties	4
	Establish requirements, scopes, plan tasks, and meet objectives through effective teamwork.	5

7. **Topics to be Covered:**

- Introduction to Engineering Design
- Conceptual Design Process
- Product Development Cycle
- Program specific topics
- Project Report and Final Presentation

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1. **Course number and name:** EE 492 Cooperative Training
2. **Credits and contact hours:** 10 (0)
3. **Instructor:** Dr. Moustafa Nasralla
4. **Textbook:** NA
5. **Course Information**
 - a. **Description:** Cooperative Education is a planned learning experience, for which credits are earned, that integrates classroom theory and learning experiences at a workplace. The experience enables students to apply and refine the knowledge and skills acquired in a related curriculum course. The Co-op is a supervised work experience in a position in the government or the private sector which aimed to assist students in making the transition from the classroom to industry. Students are required to finish 28 weeks on the job to earn 10-credit hours for the course
 - b. **Prerequisite or Co-requisite:** Completion of 128 Credit Hours
 - c. This is a Required course.
6. **Goals:** Specific goals for the course
Instructional Outcomes and their mapping to Student Outcomes:

NQF Domain	Instructional Outcomes	SO
Knowledge & Understanding	Demonstrate a comprehensive understanding of their academic studies through general education and their areas of specializations.	1
	Define and recognize complex electrical engineering problems encountered in industry, utilizing theoretical knowledge acquired in academic coursework to propose innovative and effective solutions.	1
Skills	Demonstrate effective communication skills to a range of audiences.	3
	Examine advanced technical skills in electrical engineering, including design, analysis, and troubleshooting, to successfully complete assigned tasks and projects within cooperative work environments.	2, 6
Values	Demonstrate the ability to work as an individual and as a part of a multi-disciplinary team	4, 5
	Demonstrate professional and ethical responsibility	5, 7

7. **Topics to be Covered:**

- Topics are determined by the field employer

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